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# Import Required Libraries

import pandas as pd

from imblearn.over_sampling import SMOTE


# Read Dataset

df = pd.read_csv("dataset/loan_data.csv")


# -----

# Handle Missing Values (if not already done)

# -----


# Fill numerical columns with mean

numerical_columns = [

    'ApplicantIncome',

    'CoapplicantIncome',

    'LoanAmount',

    'Loan_Amount_Term'

]


for column in numerical_columns:

    df[column] = df[column].fillna(df[column].mean())


# Fill categorical columns with mode

categorical_columns = [
```

```
'Gender',
'Married',
'Dependents',
'Education',
'Self_Employed',
'Credit_History',
'Property_Area'
]
```

```
for column in categorical_columns:
```

```
    df[column] = df[column].fillna(df[column].mode()[0])
```

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# -----
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# Convert Categorical Variables to Numeric
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# -----
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```
df = pd.get_dummies(df, drop_first=True)
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# -----
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# Separate Features and Target Variable
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# -----
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X = df.drop('Loan_Status_Y', axis=1)
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```
y = df['Loan_Status_Y']
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# -----
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```
# Display Class Distribution Before SMOTE

# -----

print("Class Distribution Before SMOTE:")
print(y.value_counts())

# -----

# Apply SMOTE

# -----

smote = SMOTE(random_state=42)

X_balanced, y_balanced = smote.fit_resample(X, y)

# -----

# Display Class Distribution After SMOTE

# -----

print("\nClass Distribution After SMOTE:")
print(y_balanced.value_counts())

# -----

# Display Dataset Shape

# -----

print("\nOriginal Dataset Shape:")
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```
print(X.shape)
```

```
print("\nBalanced Dataset Shape:")
```

```
print(X_balanced.shape)
```